

01) a) $\bar{f} \circ \bar{g} \rightarrow$ es posible

$$\left. \begin{array}{l} \text{Im } \bar{g} = \mathbb{R}_0^+ \times \mathbb{R} \\ \text{Dom } \bar{f} = \mathbb{R}^2 \end{array} \right\} \text{Im } \bar{g} \subseteq \text{Dom } \bar{f}$$

$$\text{Dom } \bar{f} \circ \bar{g} = \text{Dom } \bar{g} = \mathbb{R}^2$$

• $\bar{g} \circ \bar{f} \rightarrow$ es posible

$$\left. \begin{array}{l} \text{Im } \bar{f} = \mathbb{R}^2 \\ \text{Dom } \bar{g} = \mathbb{R}^2 \end{array} \right\} \text{Im } \bar{f} \subseteq \text{Dom } \bar{g}$$

$$\text{Dom } \bar{g} \circ \bar{f} = \text{Dom } \bar{f} = \mathbb{R}^2$$

a) • $\bar{f} \circ \bar{g} = \bar{h}_1$

$$D\bar{h}_1|_{(u_0, v_0)} = D\bar{f}|_{\bar{g}(u_0, v_0)} \cdot D\bar{g}|_{(u_0, v_0)}$$

$$D\bar{h}_1 = \begin{pmatrix} y & x \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2u & 0 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2uy - x & x \\ 2u + 1 & -1 \end{pmatrix}$$

• $\bar{g} \circ \bar{f} = \bar{h}_2$

$$D\bar{h}_2|_{(x_0, y_0)} = D\bar{g}|_{\bar{f}(x_0, y_0)} \cdot D\bar{f}|_{(x_0, y_0)}$$

$$D\bar{h}_2 = \begin{pmatrix} 2u & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} y & x \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2uy & 2ux \\ -y+1 & -x-1 \end{pmatrix}$$

TP6)

$$1) b) f(x, y) = x\sqrt{y} \quad \bar{g}(u) = (u, 2-u)$$

$f \circ \bar{g}$ Para hallar $\text{Im } \bar{g}$:

$$\left. \begin{array}{l} x = u \\ y = 2 - u \end{array} \right\} \Rightarrow y = 2 - x$$

$$\left. \begin{array}{l} \text{Im } \bar{g} = \{ (x, y) \in \mathbb{R}^2 / y = 2 - x \} \\ \text{Dom } f = \{ (x, y) \in \mathbb{R}^2 / y \geq 0 \} \end{array} \right\} \text{Im } \bar{g} \not\subset \text{Dom } f$$

Podemos restringir:

$$\bar{g}^*(u) = (u, 2-u) \text{ de modo que } 2-u \geq 0 \Rightarrow u \leq 2$$

$$\bar{g}^*: (-\infty, 2] \subset \mathbb{R} \rightarrow \mathbb{R}^2 / \bar{g}^*(u) = (u, 2-u)$$

$$\text{Im } \bar{g}^* = \{ (x, y) \in \mathbb{R}^2 / y = 2 - x \wedge y \geq 0 \} \subset \text{Dom } f$$

Luego, la composición es posible

$$f \circ \bar{g}^*: (-\infty, 2] \subset \mathbb{R} \rightarrow \mathbb{R} / (f \circ \bar{g}^*)(u) = f(\bar{g}^*(u))$$

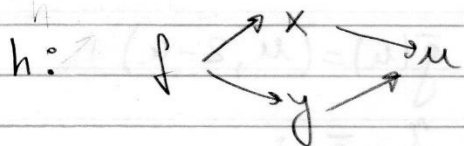
Matriz jacobiana de $f \circ \bar{g}^*$

$$\begin{aligned} D(f \circ \bar{g}^*)(u_0) &= Df(\bar{g}^*(u_0)) \cdot D\bar{g}^*(u_0) = \\ &= \nabla f(\bar{g}^*(u_0)) \cdot [\bar{g}'^*(u_0)]^T \end{aligned}$$

$$D(f \circ \bar{g}^*) = (f \circ \bar{g}^*)' = \nabla f \cdot (\bar{g}'^*)^T =$$

$$= \left(\sqrt{y} \quad \frac{x}{2\sqrt{y}} \right) \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \sqrt{y} - \frac{x}{2\sqrt{y}}$$

Uso de regla orientada para $h = f \circ \bar{g}$



Sumamos todos los "caminos" que conducen a u

$$h'(u) = f'_x \cdot x'_u + f'_y \cdot y'_u$$

$$h'(u) = \sqrt{y} \cdot 1 + \frac{x}{2\sqrt{y}} (-1)$$

$$h'(u) = \sqrt{y} - \frac{x}{2\sqrt{y}}$$

$$\bar{g}(u) = (\underbrace{u}_x, \underbrace{2-u}_y)$$

$$f(x,y) = x\sqrt{y}$$

$\bar{g} \circ f$

$$\text{Im } f = \mathbb{R}$$

$$\text{Dom } \bar{g} = \mathbb{R}$$

$$\text{Im } f \subset \text{Dom } \bar{g}$$

La composición es posible

$$\bar{g} \circ f : \underbrace{\text{Dom } f}_{\{(x,y) \in \mathbb{R}^2 / y \geq 0\}} \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2 / (\bar{g} \circ f)(x,y) = \bar{g}(f(x,y))$$

Matriz jacobiana de $\bar{g} \circ f$

$$D(\bar{g} \circ f)(x_0, y_0) = D\bar{g}(f(x_0, y_0)) \cdot Df(x_0, y_0) =$$

$$= [\bar{g}'(f(x_0, y_0))]^T \cdot \nabla f(x_0, y_0)$$

$$D(\bar{g} \circ f) = (\bar{g}')^T \cdot \nabla f = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} \sqrt{y} & \frac{x}{2\sqrt{y}} \end{pmatrix} =$$

$$= \begin{pmatrix} \sqrt{y} & \frac{x}{2\sqrt{y}} \\ -\sqrt{y} & -x/2\sqrt{y} \end{pmatrix}$$

$$1) c) \quad \bar{f}(x,y) = (x-y, \sqrt{x+y})$$

$$\bar{g}(t) = (2-t, t-3)$$

$$\bar{f} \circ \bar{g} \mid \bar{g}(t) = (2-t, t-3) \quad \left. \begin{array}{l} x = 2-t \Rightarrow t = 2-x \\ y = t-3 \end{array} \right\} \Rightarrow$$

$$\Rightarrow y = 2-x-3 \Rightarrow y = -x-1$$

$$\left. \begin{array}{l} \text{Im } \bar{g} = \{ (x,y) \in \mathbb{R}^2 / y = -x-1 \} \\ \text{Dom } \bar{f} = \{ (x,y) \in \mathbb{R}^2 / x+y \geq 0 \} \end{array} \right\} \text{Im } \bar{g} \not\subset \text{Dom } \bar{f}$$

Podemos restringir:

$$\bar{g}^*(t) = (2-t, t-3) \text{ de modo que } 2-t+t-3 \geq 0 \Rightarrow$$

$\Rightarrow -1 \geq 0$ absurdo es imposible restringir
para realizar la composición

$$\bar{g} \circ \bar{f} \mid \left. \begin{array}{l} \text{Im } \bar{f} \subset \mathbb{R}^2 \\ \text{Dom } \bar{g} = \mathbb{R} \end{array} \right\} \text{Im } \bar{f} \not\subset \text{Dom } \bar{g}$$

En este caso no puede plantearse una restricción
por incompatibilidad en las dimensiones de
 $\text{Im } \bar{f}$ y $\text{Dom } \bar{g}$

1d) • $\bar{f} \circ \bar{g} \rightarrow$ es posible

$$\left. \begin{array}{l} \text{Im } \bar{g} = \mathbb{R}^2 \\ \text{Dom } \bar{f} = \mathbb{R}^2 \end{array} \right\} \text{Im } \bar{g} \subseteq \text{Dom } \bar{f}$$

$$\text{Dom } \bar{f} \circ \bar{g} = \text{Dom } \bar{g} = \{(u, v, w) \in \mathbb{R}^3 / 1-u \geq 0\}$$

• $\bar{g} \circ \bar{f} \rightarrow$ no es posible

$$\left. \begin{array}{l} \text{Im } \bar{f} = \mathbb{R}^3 \\ \text{Dom } \bar{g} = \{(u, v, w) \in \mathbb{R}^3 / u \leq 1\} \end{array} \right\} \text{Im } \bar{f} \not\subseteq \text{Dom } \bar{g}$$

Podemos restringir

$$\bar{f}^*(x, y) = (xy^2, y-x, x) \text{ siendo } xy^2 \leq 1$$

$$\text{Dom } \bar{f}^* = \{(x, y) \in \mathbb{R}^2 / xy^2 \leq 1\}$$

$$\left. \begin{array}{l} \text{Im } \bar{f}^* = \{(u, v, w) \in \mathbb{R}^3 / u \leq 1\} \\ \text{Dom } \bar{g} = \text{idem} \end{array} \right\} \text{Im } \bar{f}^* \subseteq \text{Dom } \bar{g}$$

y por lo tanto $\bar{g} \circ \bar{f}^*$ es posible

$$\text{Dom } \bar{g} \circ \bar{f}^* = \text{Dom } \bar{f}^*$$

$$b) \bar{f} \circ \bar{g} = \bar{h}$$

$$D\bar{h}(u, v, w) = D\bar{f}|_{\bar{g}(u, v, w)} \cdot D\bar{g}(u, v, w)$$

$$D\bar{h} = \begin{pmatrix} y^2 & 2xy \\ -1 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ \frac{-w}{2\sqrt{1-u}} & 0 & \sqrt{1-u} \end{pmatrix} =$$

$$= \begin{pmatrix} y^2 - \frac{xyw}{\sqrt{1-u}} & -y^2 & 2xy\sqrt{1-u} \\ -1 - \frac{w}{2\sqrt{1-u}} & 1 & \sqrt{1-u} \\ 1 & -1 & 0 \end{pmatrix}$$

$$d) \bar{g} \circ \bar{f}^* = \bar{m}$$

$$D\bar{m}(x, y) = D(\bar{g} \circ \bar{f}^*)(x, y) = D\bar{g}|_{\bar{f}^*(x, y)} \cdot D\bar{f}^*(x, y)$$

$$D\bar{m} = \begin{pmatrix} 1 & -1 & 0 \\ \frac{-1}{2\sqrt{1-u}} & 0 & \sqrt{1-u} \end{pmatrix} \begin{pmatrix} y^2 & 2xy \\ -1 & 1 \\ 1 & 0 \end{pmatrix} =$$

$$= \begin{pmatrix} y^2 + 1 & 2xy - 1 \\ \frac{-y^2}{2\sqrt{1-u}} + \sqrt{1-u} & \frac{-xy}{\sqrt{1-u}} \end{pmatrix}$$

$$2) \quad h(x, y) = x \ln(1 - xy) \quad D = \{ \bar{x} \in \mathbb{R}^2 \mid 1 - xy > 0 \}$$

$$f: D \rightarrow \mathbb{R} \mid f(x, y) = x \ln(1 - xy)$$

$$g: \mathbb{R} \rightarrow \mathbb{R} \mid g(u) = u$$

$$h = (g \circ f)$$

$$\begin{aligned} h: D \rightarrow \mathbb{R} \mid h(x, y) &= (g \circ f)(x, y) = g(f(x, y)) = \\ &= g(x \ln(1 - xy)) = x \ln(1 - xy) \end{aligned}$$

3)

a) $\bar{g}(x, y) = (x - y^2, x + 2xy - 1)$

$\bar{g}: A \subset \mathbb{R}^2 \rightarrow \mathbb{R}^2 / \bar{g}(x, y) = (x - y^2, x + 2xy - 1)$

$f: \mathbb{R}^2 \rightarrow \mathbb{R} / f(u, v) = 2uv - 2\sqrt{v-u}$

Necesitamos $v - u \geq 0 \Rightarrow x + 2xy - 1 - x + y^2 \geq 0$
 $\Rightarrow y^2 + 2xy - 1 \geq 0$

$A = \{(x, y) \in \mathbb{R}^2 / y^2 + 2xy - 1 \geq 0\}$

b) $\vec{r} = (5, 5) - (2, 1) = (3, 4)$

$\vec{r} = \frac{(3, 4)}{\sqrt{9+16}} = \left(\frac{3}{5}, \frac{4}{5}\right)$

$\nabla h(2, 1) = \nabla(f \circ \bar{g})(2, 1) = \nabla f|_{\bar{g}(2, 1)} \cdot \nabla \bar{g}(2, 1) =$

$= \left(2v - \frac{2(-1)}{2\sqrt{v-u}}, 2u - \frac{2}{2\sqrt{v-u}} \right)_{\left(\underbrace{2-1}_1, \underbrace{2+4-1}_5 \right)}$

$\cdot \begin{pmatrix} 1 & +2y \\ 1+2y & 2x \end{pmatrix}_{(2, 1)} = \left(10 + \frac{1}{\sqrt{5-1}}, 2 - \frac{1}{\sqrt{4}} \right)$

$\cdot \begin{pmatrix} 1 & -2 \\ 1+2 & 4 \end{pmatrix} = \left(\frac{21}{2}, \frac{3}{2} \right) \begin{pmatrix} 1 & -2 \\ 3 & 4 \end{pmatrix} = \left(\frac{21}{2} + \frac{9}{2}, \frac{-42}{2} + \right.$

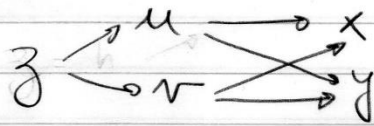
$\left. + \frac{12}{2} \right) = \left(\frac{30}{2}, -\frac{30}{2} \right) = (15, -15)$

$\frac{\partial h}{\partial \vec{r}}(2, 1) = \nabla h(2, 1) \cdot \vec{r} = (15, -15) \left(\frac{3}{5}, \frac{4}{5} \right) =$

$= \frac{15 \cdot 3}{5} - \frac{4 \cdot 15}{5} = 9 - 12 = -3$

cálculo de $\nabla h(2,1)$

3) b) trabajando con red orientada



$$x = 2$$

$$y = 1$$

$$u = 2 - 1^2 = 1$$

$$v = 2 + 2 \cdot 2 \cdot 1 - 1 = 5$$

$$h'_x = z'_u \cdot u'_x + z'_v \cdot v'_x$$

$$h'_x = \left(2v + \frac{2}{2\sqrt{v-u}} \right) \cdot 1 + \left(2u - \frac{2}{2\sqrt{v-u}} \right) \cdot (1+2y)$$

$$h'_x(2,1) = \left(2 \cdot 5 + \frac{1}{\sqrt{5-1}} \right) \cdot 1 + \left(2 \cdot 1 - \frac{1}{\sqrt{5-1}} \right) (1+2 \cdot 1)$$

$$h'_x(2,1) = 15$$

$$h'_y = z'_u \cdot u'_y + z'_v \cdot v'_y$$

$$h'_y = \left(2v + \frac{1}{\sqrt{v-u}} \right) \cdot (-2y) + \left(2u - \frac{1}{\sqrt{v-u}} \right) \cdot 2x$$

$$h'_y(2,1) = \left(2 \cdot 5 + \frac{1}{2} \right) (-2 \cdot 1) + \left(2 \cdot 1 - \frac{1}{2} \right) \cdot 2 \cdot 2$$

$$h'_y(2,1) = -15$$

$$\boxed{\nabla h(2,1) = (15, -15)}$$

c) Para hallar la recta normal halla la ecuación del plano tangente, cuyo vector director es el vector director de la recta η_0

$$z - z_0 = h'_x(2,1)(x-2) + h'_y(2,1)(y-1)$$

$$z_0 = h(2,1) = (f \circ g)(2,1) = 2 \cdot 1 \cdot 5 - 2 \cdot \sqrt{5-1} = 6$$

$$z-6 = 15(x-2) - 15(y-1)$$

$$9 = 15x - 15y - 3$$

$$\vec{n} = (15, -15, -1)$$

$$\pi_0 : (x, y, z) = (2, 1, 6) + \lambda(15, -15, -1) \quad \lambda \in \mathbb{R}$$

$$\frac{x-2}{15} = \frac{y-1}{-15} = \frac{z-6}{-1}$$

$$\left\{ \begin{array}{l} \frac{x-2}{15} = \frac{y-1}{-15} \Rightarrow x-2 = 1-y \Rightarrow x+y=3 \end{array} \right.$$

$$\left\{ \begin{array}{l} \frac{x-2}{15} = \frac{z-6}{-1} \Rightarrow x-2 = 15(6-z) \Rightarrow x+15z=92 \end{array} \right.$$

$$\boxed{\begin{cases} x+y=3 \\ x+15z=92 \end{cases}}$$

c) Ecuación eje $z \rightarrow \begin{cases} x=0 \\ y=0 \end{cases}$

Puede, para que haya intersección debe verificarse que:

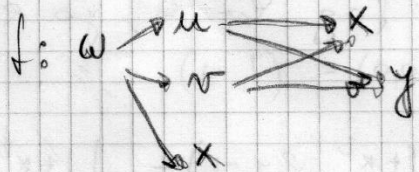
$$\begin{cases} x = 2 + 15\lambda = 0 \rightarrow \lambda = -2/15 \\ y = 1 - 15\lambda = 0 \rightarrow \lambda = 1/15 \end{cases} \text{ absurdo}$$

Reemplazando en la ecuación conteniendo:

$$x+y=3 \rightarrow 0+0=3 \text{ absurdo}$$

No hay intersección entre π_0 y el eje z

$$4) \omega = u^3 - x v^2 \quad \begin{cases} u = x \sqrt{y-x} \\ v = 2x + y^2 \end{cases}$$



$$x=0$$

$$y=1$$

$$u=0$$

$$v=1^2=1$$

$$f'_x = w'_x + w'_u u'_x + w'_v v'_x = -v^2 + 3u^2 \cdot \left(\sqrt{y-x} - \frac{x}{2\sqrt{y-x}} \right) + (-2xv) \cdot 2$$

$$f'_x = -v^2 + 3u^2 \sqrt{y-x} - \frac{3u^2 x}{2\sqrt{y-x}} - 4xv$$

$$f'_x(0,1) = -1 + 0 - 0 - 0 = -1$$

05) $g(0,0) = (1, -1)$

$$h(0,0) = f(g(0,0)) = f(1, -1) = |1-1| + 1 = 1$$

No puede aplicarse la regla de la cadena para calcular las derivadas de $h = f \circ g$ en $\vec{0}$, ya que f no es diferenciable en $g(0,0) = (1, -1)$

$$h = (f \circ g)(x,y) = f(g(x,y)) = f(1+x^2, 2y-1) = |1+x^2-1| - 2y+1 = |x^2| - 2y+1 = |x|^2 - 2y+1 = x^2 - 2y+1$$

$$\begin{cases} h'_x = 2x \\ h'_y = -2 \end{cases} \Rightarrow \begin{cases} h'_x(0,0) = 0 \\ h'_y(0,0) = -2 \end{cases}$$

f no es dif en $g(0,0) = (1, -1)$ porque $\nexists \frac{\partial f}{\partial u}(1, -1)$

$$\begin{aligned} \frac{\partial f}{\partial u}(1, -1) &= \lim_{h \rightarrow 0} \frac{f(1+h, -1) - f(1, -1)}{h} = \lim_{h \rightarrow 0} \frac{|1+h-1| + 1 - 1}{h} = \\ &= \lim_{h \rightarrow 0} \frac{|h|}{h} \rightarrow \begin{cases} \lim_{h \rightarrow 0^+} \frac{h}{h} = 1 \\ \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1 \end{cases} \Rightarrow \nexists \frac{\partial f}{\partial u}(1, -1) \end{aligned}$$

NOTA

$$6) \frac{\partial z}{\partial x} = f' \cdot \left(\frac{2x(x^2+y^2) - 2x(x^2-y^2)}{(x^2+y^2)^2} \right) =$$

$$= f' \cdot \frac{\cancel{2x^3} + 2xy^2 - \cancel{2x^3} + 2xy^2}{(x^2+y^2)^2} = f' \cdot \frac{4xy^2}{(x^2+y^2)^2}$$

$$\frac{\partial z}{\partial y} = f' \cdot \left(\frac{-2y(x^2+y^2) - 2y(x^2-y^2)}{(x^2+y^2)^2} \right) =$$

$$= f' \cdot \frac{-\cancel{2yx^2} - \cancel{2y^3} - \cancel{2yx^2} + \cancel{2y^3}}{(x^2+y^2)^2} = \frac{-4yx^2}{(x^2+y^2)^2}$$

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{4x^2y^2}{(x^2+y^2)^2} - \frac{4y^2x^2}{(x^2+y^2)^2} = 0$$

$$7) h(x,y) = f\left(\frac{2x}{y}\right) - f\left(\frac{y}{x}\right) \quad f \in C^1$$

$$h'_x = f' \cdot \frac{2}{y} - f' \left(-\frac{y}{x^2}\right)$$

$$h'_y = f' \left(-\frac{2x}{y^2}\right) - f' \left(\frac{1}{x}\right)$$

$$x \cdot h'_x + y h'_y = x \left(f' \cdot \frac{2}{y} + f' \frac{y}{x^2} \right) + y \left(-f' \frac{2x}{y^2} - f' \cdot \frac{1}{x} \right) = \cancel{\frac{2x}{y} f'} + \cancel{\frac{y}{x} f'} - \cancel{\frac{2x}{y} f'} - \cancel{\frac{y}{x} f'} = 0$$

$$8) \quad \nabla h(1,1) = \nabla f(\bar{g}(1,1)) \cdot D\bar{g}(1,1) =$$

$$= (2x, 2y)|_{(2,a)} \cdot \begin{pmatrix} 1 & 1 \\ 0 & a \end{pmatrix} = (4, 2a) \begin{pmatrix} 1 & 1 \\ 0 & a \end{pmatrix} =$$

$$= (4, 4 + 2a^2)$$

(conk > 0)

$$\bar{r} = (5, 7) = k(4, 4 + 2a^2) \Rightarrow \begin{cases} 5 = 4k \\ 7 = (4 + 2a^2)k \end{cases}$$

$$k = 5/4$$

$$7 = (4 + 2a^2) \frac{5}{4} \Rightarrow 28 = 20 + 10a^2 \Rightarrow \frac{8}{10} = a^2 \Rightarrow$$

$$\Rightarrow a = \pm \frac{2}{\sqrt{5}}$$

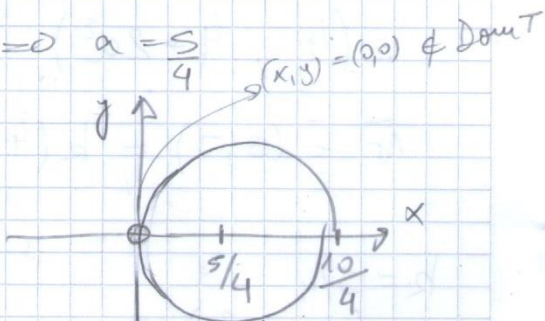
9) a) $T(x, y) = \frac{3x}{x^2 + y^2} = T(2, -1) = \frac{6}{4+1}$

$\frac{\partial T}{\partial x} = \frac{3}{5} \Rightarrow \frac{5}{2}x = x^2 + y^2$ (Kreis
desplazado)

$x^2 - \frac{5}{2}x + \frac{25}{16} + y^2 = \frac{25}{16}$

$2 \cdot x \cdot a = \frac{5}{2}x \Rightarrow 2a = \frac{5}{2} \Rightarrow a = \frac{5}{4}$

$(x - \frac{5}{4})^2 + y^2 = (\frac{5}{4})^2$



b) $\nabla T = \left(\frac{3(x^2 + y^2) - 2x \cdot 3x}{(x^2 + y^2)^2}, \frac{-2y \cdot 3x}{(x^2 + y^2)^2} \right) =$
 $= \left(\frac{-3x^2 + 3y^2}{(x^2 + y^2)^2}, \frac{-6xy}{(x^2 + y^2)^2} \right)$

$\nabla T(2, -1) = \left(\frac{-3 \cdot 4 + 3 \cdot 1}{25}, \frac{-6 \cdot 2 \cdot (-1)}{25} \right) = \left(\frac{-9}{25}, \frac{12}{25} \right)$

$\|\nabla T(2, -1)\| = \sqrt{\frac{81}{25^2} + \frac{144}{25^2}} = \sqrt{\frac{225}{25}} = \frac{15}{25} = \frac{3}{5}$

$\vec{v} = \left(\frac{-9/25}{3/5}, \frac{12/25}{3/5} \right) = \left(-\frac{3}{5}, \frac{4}{5} \right)$

d) $T \begin{matrix} \nearrow x \\ \searrow y \end{matrix} \begin{matrix} \nearrow t \\ \searrow \end{matrix}$

$T'_t = T'_x \cdot x'_t + T'_y \cdot y'_t$

$$T'(t) = \frac{-3x^2 + 3y^2}{(x^2 + y^2)^2} \cdot 2 \cos t + \frac{-6xy}{(x^2 + y^2)^2} \cdot (-\sin t) =$$

$$= \frac{-3 \cdot 4 \sin^2 t + 3 \cos^2 t}{(4 \sin^2 t + \cos^2 t)^2} \cdot 2 \cos t -$$

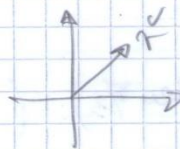
$$- \frac{6 \cdot 2 \sin t \cos t}{(4 \sin^2 t + \cos^2 t)^2} \cdot (-\sin t) =$$

$$= \frac{-24 \sin^2 t \cos t + 6 \cos^3 t + 12 \sin^2 t \cos t}{(3 \sin^2 t + \underbrace{\sin^2 t + \cos^2 t}_{=1})^2} =$$

$$= \frac{6 \cos t (-2 \sin^2 t + \cos^2 t)}{(3 \sin^2 t + 1)^2} =$$

$$= \frac{6 \cos t (-3 \sin^2 t + \underbrace{\sin^2 t + \cos^2 t}_{=1})}{(3 \sin^2 t + 1)^2} =$$

$$= \frac{6 \cos t (1 - 3 \sin^2 t)}{(3 \sin^2 t + 1)^2}$$

c)  $\vec{r} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$ $\nabla T(2, -1) = \left(-\frac{9}{25}, \frac{12}{25} \right)$

$$T'(\vec{A}, \vec{r}) = \left(-\frac{9}{25}, \frac{12}{25} \right) \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) = \frac{-9 + 12}{25\sqrt{2}} =$$

$$= \left(\frac{3}{25\sqrt{2}} \right)$$

$$10) \quad xy - e^{z-x} = \ln z \rightarrow G(x,y,z) = xy - e^{z-x} - \ln z = 0$$

$G(x,y,z)=0$ define implícitamente $z=f(x,y)$

Expresión lineal que aproxime la función $f \rightarrow$ plano tangente $z_P \cong f(x,y)$ en $E(1,1)$

$$z_P - z_0 = f'_x(1,1)(x-1) + f'_y(1,1)(y-1)$$

$$f'_x = - \frac{G'_x}{G'_z}$$

$$f'_x(1,1) = - \frac{G'_x(1,1,1)}{G'_z(1,1,1)}$$

$$f'_x(1,1) = - \frac{2}{-2} = 1$$

$$f'_y = - \frac{G'_y}{G'_z}$$

$$f'_y(1,1) = - \frac{G'_y(1,1,1)}{G'_z(1,1,1)} = - \frac{1}{-2} = \frac{1}{2}$$

$$z_P - 1 = 1(x-1) + \frac{1}{2}(y-1) \Rightarrow z_P = x + \frac{1}{2}y - \frac{1}{2}$$

$$\boxed{f(x,y) \cong x + \frac{1}{2}y - \frac{1}{2} \text{ en } E(1,1)}$$

Calc aux

$$G(x,y,z) = xy - e^{z-x} - \ln z = 0$$

$$G(1,1,z_0) = 1 \cdot 1 - e^{z_0-1} - \ln z_0 = 0 \Rightarrow z_0 = 1$$

$$G'_x = y + e^{z-x} \Rightarrow G'_x(1,1,1) = 1+1=2$$

$$G'_y = x \Rightarrow G'_y(1,1,1) = 1$$

$$G'_z = -e^{z-x} - \frac{1}{z} \Rightarrow G'_z(1,1,1) = -2$$

$$11) a) z^3 + 2xz + yz - x = 0 = G(x, y, z)$$

$$f'_x = - \frac{G'_x}{G'_z} = - \frac{2z - 1}{3z^2 + 2x + y}$$

$$f'_y = - \frac{G'_y}{G'_z} = - \frac{z}{3z^2 + 2x + y}$$

$$z_0^3 + 2 \cdot 1 \cdot z_0 + (-2)z_0 - 1 = 0$$

$$z_0^3 = 1 \Rightarrow z_0 = 1$$

$$x_0 = 1$$

$$y_0 = -2$$

$$z_0 = 1$$

$$f'_x(1, -2) = - \frac{2 \cdot 1 - 1}{3 \cdot 1 + 2 - 2} = - \frac{1}{3}$$

$$f'_y(1, -2) = - \frac{1}{3}$$

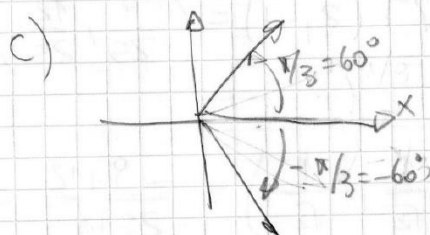
$$\nabla f(1, -2) = (-1/3, -1/3)$$

b) Planus Tangente $z - 1 = -\frac{1}{3}(x - 1) - \frac{1}{3}(y + 2)$

$$3z - 3 = -x + 1 - y - 2 \Rightarrow 3z + x + y = 2$$

Rechte normal $\vec{m} = (1, 1, 3)$

$$(x, y, z) = (1, -2, 1) + \lambda(1, 1, 3) \quad \lambda \in \mathbb{R}$$



$$\vec{r}_1 = (\cos \pi/3, \sin \pi/3) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

$$\vec{r}_2 = (\cos(-\pi/3), \sin(-\pi/3)) = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

$$\frac{\partial f}{\partial \vec{r}_1}(1, -2) = \nabla f(1, -2) \cdot \vec{r}_1 = \left(-\frac{1}{3}, -\frac{1}{3}\right) \cdot \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) = \frac{-1 - \sqrt{3}}{6}$$

$$\frac{\partial f}{\partial \vec{r}_2}(1, -2) = \nabla f(1, -2) \cdot \vec{r}_2 = \left(-\frac{1}{3}, -\frac{1}{3}\right) \cdot \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right) = \frac{-1 + \sqrt{3}}{6}$$

$$12) F(x, y, z) = 2x^2 - 2y^2 + z^2 - 1 = 0$$

$$\nabla F = (4x, -4y, 2z)$$

Plano tangente en (x_0, y_0, z_0)

$$(4x_0, -4y_0, 2z_0)(x - x_0, y - y_0, z - z_0) = 0 \Rightarrow$$

$$\Rightarrow 4x_0(x - x_0) - 4y_0(y - y_0) + 2z_0(z - z_0) = 0$$

Para que este plano sea paralelo al $\pi: z = x - y \Rightarrow$

$$\Leftrightarrow x - y - z = 0 \text{ no debe dar } 0$$

$$\left. \begin{array}{l} 4x_0 = 1 \cdot k \\ -4y_0 = -1 \cdot k \\ 2z_0 = -1 \cdot k \end{array} \right\} \Rightarrow \left. \begin{array}{l} x_0 = k/4 \\ y_0 = k/4 \\ z_0 = -k/2 \end{array} \right\} \quad (\text{con } k \in \mathbb{R})$$

Verifico que pts pertenecen al hiperboloide

$$\frac{2k^2}{16} - 2\frac{k^2}{16} + \frac{k^2}{4} = 1 \Rightarrow k^2 = 4 \Rightarrow k = 2 \vee k = -2$$

$$k = 2 \rightarrow P = \left(\frac{1}{2}, \frac{1}{2}, -1\right)$$

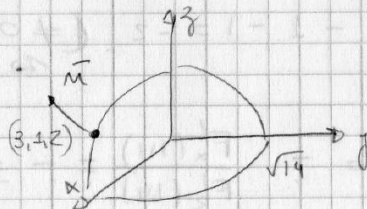
$$k = -2 \rightarrow P = \left(-\frac{1}{2}, -\frac{1}{2}, 1\right)$$

$$13) G(x, y, z) = x^2 + y^2 + z^2 - 14 = 0$$

$$\nabla G = (2x, 2y, 2z)$$

$$\nabla G(3, 1, 2) = (6, 2, 4)$$

Verifico si $\vec{u} = (6, 2, 4)$ es interior



$\vec{u}$ es exterior, por lo tanto $\vec{r} = -\vec{u} = (-6, -2, -4)$ es interior

$$\frac{\partial f}{\partial x}(3, 1, 2) = \nabla f(3, 1, 2) \cdot \vec{r} =$$

$$= (16x - 2y, -2x, 0) \Big|_{(3, 1, 2)} \cdot \frac{(-6, -2, -4)}{\sqrt{36 + 4 + 16}} =$$

$$= (46, -6, 0) \cdot \frac{(-6, -2, -4)}{\sqrt{44}} = \frac{-276 + 12}{\sqrt{44}} = \frac{-264}{\sqrt{44}} = \frac{-132}{\sqrt{11}}$$

14) líneas de nivel $\rightarrow \ln(x^2+y^2) = \ln(x_0^2+y_0^2) \Rightarrow$
 $\Rightarrow x^2+y^2 = x_0^2+y_0^2 = a^2 \rightarrow$ circunf con centro en el
origen y radio $\sqrt{x_0^2+y_0^2}$

$$G(x,y) = x^2+y^2 - a^2 = 0$$

$$\nabla G = (2x, 2y) \quad \nabla G(x_0, y_0) = (2x_0, 2y_0)$$

$$\vec{F} = (2x_0, 2y_0) \quad \vec{r} = \frac{(2x_0, 2y_0)}{\sqrt{4x_0^2+4y_0^2}} = \frac{(x_0, y_0)}{\sqrt{x_0^2+y_0^2}}$$

$$\frac{\partial f}{\partial \vec{r}}(x_0, y_0) = \nabla f(x_0, y_0) \cdot \vec{r} = \left(\frac{2x}{x^2+y^2}, \frac{2y}{x^2+y^2} \right)_{(x_0, y_0)} \cdot \vec{r} =$$

$$= \left(\frac{2x_0}{x_0^2+y_0^2}, \frac{2y_0}{x_0^2+y_0^2} \right) \left(\frac{x_0, y_0}{\sqrt{x_0^2+y_0^2}} \right) =$$

$$= \frac{2x_0^2+2y_0^2}{(x_0^2+y_0^2)\sqrt{x_0^2+y_0^2}} = \frac{2}{\sqrt{x_0^2+y_0^2}}$$

15) $\bar{A} \rightarrow$ pto de una sup de nivel de f (S es la sup)

$\pi_0 \rightarrow$ plano tge a S en $\bar{A}$

$$S \rightarrow f(x, y, z) = k$$

$$\bar{x} \in \pi_0 \quad \wedge \quad \bar{r} = \overrightarrow{A\bar{x}}$$

$$\frac{\partial f}{\partial \bar{r}}(\bar{A}) = 0 \quad \forall \bar{A}, \forall \bar{x}$$

S admite plano tge en $\bar{A} = (a_1, a_2, a_3) \Rightarrow f$ es diferenciable en $\bar{A}$

$$\frac{\partial f}{\partial \bar{r}}(\bar{A}) = \nabla f(\bar{A}) \cdot \bar{r}$$

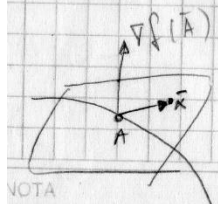
$$\text{Si tenemos que: } \bar{r} = \overrightarrow{A\bar{x}} \Rightarrow \bar{r} = \frac{\overrightarrow{A\bar{x}}}{|\overrightarrow{A\bar{x}}|}$$

$$\frac{\partial f}{\partial \bar{A}\bar{x}}(\bar{A}) = \nabla f(\bar{A}) \cdot \frac{\overrightarrow{A\bar{x}}}{|\overrightarrow{A\bar{x}}|} \in \pi_0 = 0$$

porque $\nabla f(\bar{A}) \perp \overrightarrow{A\bar{x}}$

ya que el grad es

perpendicular a la sup de nivel, es decir a su plano tge



$\perp \pi_0$

$$16) \begin{cases} x^2 - y^2 = 12 \\ z = x + y^2 \end{cases} \Rightarrow y^2 = x^2 - 12$$

$$z = x + x^2 - 12$$

$$y^2 = x^2 - 12 \quad \text{si } y > 0 \Rightarrow y = \sqrt{x^2 - 12} \quad (\text{vale porque necesitamos la curva en } E(4, 2, 8))$$

Para poder aplicar la implícita, hacemos:

$$C: \bar{g}(x) = (x, y(x), x + x^2 - 12) \quad \text{donde } y(x) \text{ queda definida implícitamente por } x^2 - y^2 = 12 \quad (\text{donde } x=4, y=2 \text{ verifican})$$

$$x=4 \Rightarrow \bar{g}(4) = (4, 2, 4 + 16 - 12) = (4, 2, 8)$$

$$\bar{g}'(x) = (1, y'(x), 1 + 2x)$$

$$y' \rightarrow F(x, y) = x^2 - y^2 - 12 = 0$$

$$y' = -\frac{F'_x}{F'_y} = -\frac{2x}{-2y} = +\frac{x}{y} \Rightarrow y'(4) = \frac{4}{2} = +2$$

$$\text{Por lo tanto: } \bar{g}'(4) = (1, +2, 1+8) = (1, +2, 9)$$

Prueba de la recta

$$(x, y, z) = (4, 2, 8) + t(1, +2, 9)$$

$$\begin{cases} x = 4+t \\ y = 2+2t \\ z = 8+9t \end{cases}$$

Si la recta corta al cilindro $y = x^2$, debe verificarse para algún valor de t que: $\frac{2+2t}{y} = \frac{(4+t)^2}{x^2}$

$$2+2t = 16 + 8t + t^2 \Rightarrow 14 + 6t + t^2 = 0 \Rightarrow t = \frac{-6 \pm \sqrt{36 - 56}}{2}$$

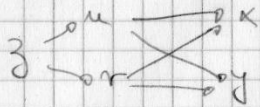
$$\Rightarrow t = \frac{-6 \pm \sqrt{-20}}{2} \Rightarrow \nexists t \in \mathbb{R} \text{ que cumpla lo pedido}$$

NOTA

NO HAY INTERSECCIÓN

$$17) z = u + r e^{u-r}$$

$$\text{con } \begin{cases} u = f(x, y) \rightarrow 2y - ux - \ln u = 0 \\ r = y^2 \end{cases}$$



$$h'(2,1, \tilde{r}) = \nabla h(2,1) \cdot \tilde{r} = 0 \quad \text{donde } \tilde{r} \perp \nabla h(2,1) \quad (A)$$

$$\text{Luego: Debo hallar } \nabla h(2,1) = [h'_x(2,1), h'_y(2,1)]$$

$$\bullet h'_x = z'_x = z'_u u'_x + z'_r r'_x = (1 + r e^{u-r}) \cdot u'_x + (e^{u-r} - r e^{u-r}) \cdot 0 = (1 + r e^{u-r}) u'_x$$

Cálculo de u'_x

$$F(x, y, u) = 2y - ux - \ln u = 0$$

$$u'_x = - \frac{F'_x}{F'_u} = - \frac{-u}{-x - \frac{1}{u}} = \frac{-u}{\frac{xu+1}{u}} = \frac{-u^2}{xu+1}$$

$$h'_x = (1 + r e^{u-r}) \left(\frac{-u^2}{xu+1} \right)$$

$$\left. \begin{matrix} x=2 \\ y=1 \end{matrix} \right\} = 0 \quad \left. \begin{matrix} 2 \cdot 1 - u \cdot 2 - \ln u = 0 \\ r = 1 \end{matrix} \right\} = 0 \quad \left. \begin{matrix} 2u + \ln u = 2 \\ r = 1 \end{matrix} \right\} = 0$$

$$\Rightarrow \begin{matrix} u=1 \\ r=1 \end{matrix}$$

$$h'_x(2,1) = (1 + 1 e^0) \left(\frac{-1}{2+1} \right) = 2 \left(\frac{-1}{3} \right) = -\frac{2}{3}$$

$$\bullet h'_y = z'_y = z'_u u'_y + z'_r r'_y = (1 + r e^{u-r}) u'_y + (e^{u-r} - r e^{u-r}) \cdot 2y$$

Cálculo u'_y

$$u'_y = - \frac{F'_y}{F'_u} = - \frac{2}{-x - \frac{1}{u}} = \frac{2u}{xu+1}$$

$$h'_y = (1 + r e^{u-r}) \frac{2u}{ru+1} + (e^{u-r} - r e^{u-r}) \cdot 2y$$

$$h'_y(2,1) = (1 + 1e^0) \cdot \frac{2}{2+1} + (1e^0 - 1e^0) \cdot 2 =$$

$$= 2 \cdot \frac{2}{3} + 0 = \frac{4}{3}$$

$$\nabla h(2,1) = \left(-\frac{2}{3}, \frac{4}{3} \right)$$

Por lo expresado en (A) :

$$\bar{r}_1 = \left(\frac{4}{3}, \frac{2}{3} \right) \Rightarrow \check{r}_1 = \frac{\left(\frac{4}{3}, \frac{2}{3} \right)}{\sqrt{\frac{16+4}{9}}} = \frac{(4,2)}{\sqrt{20}} = \frac{(4,2)}{2\sqrt{5}} = \left(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}} \right)$$

$$\bar{r}_2 = \left(-\frac{4}{3}, -\frac{2}{3} \right) \Rightarrow \check{r}_2 = \left(-\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}} \right)$$

$$18) \quad z = xy - 3x$$

$$\text{may en } z=0 \Rightarrow y = x^2 - 3$$

$$\begin{cases} z = xy - 3x \\ y = x^2 - 3 \end{cases} \Rightarrow z = x(x^2 - 3) - 3x = x^3 - 3x - 3x = x^3 - 6x$$

$$\vec{g}(x) = (x, x^2 - 3, x^3 - 6x)$$

$$\vec{g}(x_0) = (2, 1, -4) \Rightarrow \begin{cases} x_0 = 2 \\ x_0^2 - 3 = 1 \\ x_0^3 - 6x_0 = -4 \end{cases} \Rightarrow \begin{cases} x_0 = 2 \\ 4 - 3 = 1 \text{ verifica} \\ 8 - 12 = -4 \text{ verifica} \end{cases}$$

$$\vec{g}(2) = \vec{A} = (2, 1, -4)$$

$$\vec{g}'(x) = (1, 2x, 3x^2 - 6) \Rightarrow \vec{g}'(2) = (1, 4, 6)$$

Plano normal

$$(x-2, y-1, z+4) \cdot (1, 4, 6) = 0$$

$$x-2 + 4(y-1) + 6(z+4) = 0 \Rightarrow x-2 + 4y-4 + 6z+24 = 0 \Rightarrow$$

$$\Rightarrow \boxed{x + 4y + 6z + 18 = 0}$$

19) $y = h = \log \bar{g}$

$\begin{matrix} \nearrow u \\ \searrow r \\ \bar{g} = e^x \end{matrix}$

$$y' = h' = (\log \bar{g})' = \nabla f \circ \bar{g}' = (f'_u, f'_r) \cdot (e^x, e^{x^2})^T$$

Cálculo ∇f

$$F(u, r, y) = y - 1 + \ln(yur) = 0$$

$$y'_u = -\frac{F'_u}{F'_y} = -\frac{\frac{y r}{y u r}}{1 + \frac{u r}{y u r}} = \frac{-\frac{1}{u}}{\frac{y+1}{y}} = -\frac{1}{u} \cdot \frac{y}{y+1} = -\frac{y}{u(y+1)}$$

$$y'_r = -\frac{F'_r}{F'_y} = -\frac{\frac{y u}{y u r}}{1 + \frac{u r}{y u r}} = \frac{-\frac{1}{r}}{\frac{y+1}{y}} = -\frac{1}{r} \cdot \frac{y}{y+1} = -\frac{y}{r(y+1)}$$

$$y' = \left[\frac{-y}{u(y+1)}, \frac{-y}{r(y+1)} \right] [e^x, 2x e^{x^2}] =$$

$$y' = \frac{-y e^x}{u(y+1)} - \frac{2x y e^{x^2}}{r(y+1)}$$

En la composición $\rightarrow \begin{cases} u = e^x \\ r = e^{x^2} \end{cases}$

$$(1+y) y' + (1+2x) y = 0$$

$$= 0 \quad (1+y) \left[\frac{-y r e^x - 2x y u e^{x^2}}{u r (y+1)} \right] + (1+2x) y = 0$$

$$= 0 \quad \frac{-y e^{x^2} e^x - 2x y e^x e^{x^2}}{e^x e^{x^2}} + (1+2x) y = 0$$

$$= 0 \quad -y - 2xy + y + 2xy = 0$$

satisface
la ecuación
plantada

$$20) \quad x + yz - e^z = 0 = F(x, y, z)$$

$$z'_x = - \frac{F'_x}{F'_z} = - \frac{1}{y - e^z} = \frac{1}{e^z - y}$$

$$z'_y = - \frac{F'_y}{F'_z} = - \frac{z}{y - e^z} = \frac{z}{e^z - y}$$

$$z \cdot z'_x - z'_y = z \cdot \frac{1}{e^z - y} - \frac{z}{e^z - y} = 0 \quad \text{Verifizierte}$$

$$21) \quad h'_{\max}(1,1) = \|\nabla h(1,1)\|$$

$$\nabla h(1,1) = \nabla f[\bar{g}(1,1)] = D\bar{g}(1,1) \quad (A)$$

$$\bullet \quad D\bar{g} = \begin{pmatrix} y^2 & 2xy \\ -2x & 1 \end{pmatrix} \Rightarrow D\bar{g}(1,1) = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}$$

$$\bullet \quad \nabla f = (f'_u, f'_v)$$

$$\text{Sei} \quad F(u, v, z) = z - u^2 + v^2 + \ln(v+z) = 0$$

NOTA

$$z'_u = f'_u = -\frac{F'_u}{F'_z} = -\frac{-2u}{1 + \frac{1}{r+z}} = \frac{2u}{\frac{r+z+1}{r+z}}$$

$$= \frac{2(r+z)u}{r+z+1}$$

$$z'_r = f'_r = -\frac{F'_r}{F'_z} = -\frac{2r + \frac{1}{r+z}}{\frac{r+z+1}{r+z}} =$$

$$= \frac{\frac{-2r(r+z)-1}{r+z}}{\frac{r+z+1}{r+z}} = \frac{-2r^2-2rz-1}{r+z+1}$$

$$\nabla f = \left(\frac{2(r+z)u}{r+z+1}, \frac{-2r^2-2rz-1}{r+z+1} \right)$$

$$\nabla f [g(1,1)] = \nabla f(1,0) = \left(\frac{2(0+1) \cdot 1}{0+1+1}, \frac{-0-0-1}{0+1+1} \right) = \left(1, -\frac{1}{2} \right)$$

$$\text{auxiliares} \rightarrow u=1 \wedge r=0 \rightarrow z-1+0+\ln z=0 \Rightarrow$$

$$\Rightarrow z + \ln z = 1 \Rightarrow z=1$$

$$\text{Reemplazando en (A)} \rightarrow \nabla h(1,1) = \left(1, -\frac{1}{2} \right) \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix} =$$

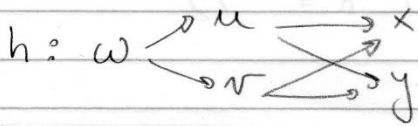
$$= \left(1 + \frac{2}{2}, 2 - \frac{1}{2} \right) = \left(2, \frac{3}{2} \right)$$

$$h'_{\max}(1,1) = \left\| \left(2, \frac{3}{2} \right) \right\| = \sqrt{4 + \frac{9}{4}} = \frac{5}{2}$$

22) Para calcular aprox $h(2,98, 2,01)$ utilizo la aprox lineal del plano tangente

$$z_p - \omega_0 = h'_x(3,2) \underbrace{(x-3)}_{-0,02} + h'_y(3,2) \underbrace{(y-2)}_{0,01}$$

$$z_p = h'_x(3,2)(-0,02) + h'_y(3,2) \cdot 0,01 + \omega_0 \quad (A)$$



$$\begin{aligned} x &= 3 \\ y &= 2 \end{aligned} \quad \begin{aligned} v &= 1 \\ u &= 3 + 2^2 = 7 \end{aligned}$$

$$\begin{aligned} M(x, y, v) &= x v + \ln(v + y - 2) - 3 = 0 \\ 3 v_0 + \ln(v_0 + 2 - 2) - 3 &= 0 \Rightarrow v_0 = 1 \end{aligned}$$

$$r'_x = -\frac{M'_x}{M'_v} = -\frac{v}{x + \frac{1}{v+y-2}} \Rightarrow r'_x(3,2) = -\frac{1}{3 + \frac{1}{1+2-2}} = -\frac{1}{4}$$

$$r'_y = -\frac{M'_y}{M'_v} = -\frac{\frac{1}{v+y-2}}{x + \frac{1}{v+y-2}} \Rightarrow r'_y(3,2) = -\frac{\frac{1}{4}}{\frac{1}{4}} = -\frac{1}{4}$$

$$h'_x = \omega'_u \cdot u'_x + \omega'_v \cdot v'_x = (2uv) \cdot 1 + (u^2 + 6v) \cdot r'_x$$

$$h'_x(3,2) = 2 \cdot 7 \cdot 1 + (49 + 6 \cdot 1) \left(-\frac{1}{4}\right) \Rightarrow h'_x(3,2) = \frac{1}{4}$$

$$h'_y = \omega'_u \cdot u'_y + \omega'_v \cdot v'_y = (2uv) \cdot 2y + (u^2 + 6v) \cdot r'_y$$

$$h'_y(3,2) = 2 \cdot 7 \cdot 4 + (49 + 6) \left(-\frac{1}{4}\right) \Rightarrow h'_y(3,2) = \frac{169}{4}$$

Retomando (A), calculamos $\omega_0 = \underbrace{7^2 \cdot 1 + 3 \cdot 1^2}_{u^2 v + 3v^2} = 52$

$$z_p = \frac{1}{4}(-0,02) + \frac{169}{4} \cdot 0,01 + 52 = 52,4175 \approx h(2,98, 2,01)$$

23) $h(t)$ decreciente en $t_0=2$ si $h'(2) < 0$

$$h(t) = f(\bar{x}(t)) = (f \circ \bar{x})(t)$$

$$h'(t) = \nabla f(\bar{x}(t)) \cdot \bar{x}'(t)$$

$$h'(2) = \nabla f(\bar{x}(2)) \cdot \bar{x}'(2) = \nabla f(1, 2) \cdot (2t, \cos(t-2))_{t=2}$$

$$h'(2) = \nabla f(1, 2) \begin{pmatrix} 4 \\ 1 \end{pmatrix}$$

$$\frac{u \cdot 2 + 1 + e^{2u+y+x-5}}{-4=0 \rightarrow u=1}$$

$$H(x, y, u) = u \cdot 2 + 1 + e^{2u+y+x-5}$$

$$u'_x = - \frac{H'_x}{H'_u} = - \frac{1 + e^{2u+y+x-5}}{y + 2e^{2u+y+x-5}}$$

$$u'_x(1, 2) = - \frac{1 + e^0}{2 + 2e^0} = - \frac{2}{4} = - \frac{1}{2}$$

$$\begin{cases} x=1 \\ y=2 \\ u=1 \end{cases}$$

$$u'_y = - \frac{H'_y}{H'_u} = - \frac{u + e^{2u+y+x-5}}{y + 2e^{2u+y+x-5}}$$

$$u'_y(1, 2) = - \frac{1 + e^0}{2 + 2e^0} = - \frac{2}{4} = - \frac{1}{2}$$

$$h'(2) = \begin{pmatrix} -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} = -2 - \frac{1}{2} < 0 \rightarrow \text{decreciente}$$

$$24) \quad h(x) = f(x, g(x), g'(x))$$

$$h \text{ cte} \rightarrow h'(x) = 0$$

$$h'(x) = \nabla f \cdot \begin{pmatrix} g(x) + x g'(x) \\ g'(x) \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} g(x) + x g'(x) \\ g'(x) \end{pmatrix} =$$

$$= g(x) + x g'(x) - g'(x) = 0$$

$$g + x g' - g' = g + (x-1) g' = 0 \Rightarrow g = (1-x) g' \Rightarrow$$

$$\Rightarrow g = (1-x) \frac{dg}{dx} \Rightarrow \int \frac{dx}{1-x} = \int \frac{dg}{g} \Rightarrow -\ln(1-x) =$$

$$= \ln(kg) \Rightarrow \ln(1-x)^{-1} = \ln(kg) \Rightarrow$$

$$\Rightarrow kg = \frac{1}{1-x} \Rightarrow g = \frac{c}{1-x}$$

$$g(3) = 1 \Rightarrow \frac{c}{1-3} = 1 \Rightarrow c = -2$$

$$g(x) = \frac{-2}{1-x} \Rightarrow \boxed{g(x) = \frac{2}{x-1}}$$

$$25) \quad y = x f(x^2-1) \text{ es solución de } xy' - y = 2x^3$$

$$y' = f(x^2-1) + x f'(x^2-1), 2x = f(x^2-1) + 2x^2 f'(x^2-1)$$

$$x [f(x^2-1) + 2x^2 f'(x^2-1)] - x f(x^2-1) = 2x^3$$

$$x \cancel{f(x^2-1)} + 2x^3 f'(x^2-1) - x \cancel{f(x^2-1)} = 2x^3$$

$$f'(\underbrace{x^2-1}_u) = 1 \Rightarrow \frac{df}{du} = 1 \Rightarrow$$

$$\Rightarrow \int df = \int du \Rightarrow \boxed{f(u) = u + c}$$